Unchanged Adam versus each intervention

Seed 0; states 673-745. Each panel has its own Y range; solid, unsmoothed trajectories.

Half learning rate

8-update cooldown

Cooldown + cosine recovery

Unchanged Adam — Intervention

Training loss (log)

ρ_t (raw adjacent)

$\|g_t\|$ (log)

$\widehat{\eta\kappa}_t$

Actual LR / base LR

Parameter state s

$\rho_t = \cos(g_t, g_{t-1})$ for raw full-objective gradients. $\widehat{\eta\kappa}_t$ threshold 1.8. LR at s applies to s -> s+1.